6 Packaging Technology

Packaging Technology

1. Packaging of the VLSI die provides mechanical and environment protection.
2. Cost and/or performance consideration of the packaged device usually dictates assembly and packaging details.
3. Packaging significantly affects the reliability of the packaged die.
4. Length of the bonding wire between the chip and package, and the lead length in the package determines the inductive voltage drop in the output circuit.
5. Good packages should provide low thermal resistance, and hence, there is limited temperature increase beyond the ambient temperature due to power dissipation.
6. Since the choice of proper packaging technology is critical to the success of the chip development, chip designers should work closely with package designers from the start of the project, especially for full custom designs.
7. Some of the important packaging concerns are as follows.
   a. Hermetic seals to prevent the penetration of moisture
   b. Thermal conductivity
   c. Thermal expansion coefficient
   d. Pin density
   e. Parasitic inductance and capacitance
   f. a-particle protection

Thermal conductivity (K) is the property of the material to conduct heat.

1. Heat should be transferred at a low rate for low K (W/cm°C) materials.
2. It should be high enough.

Thermal Coefficient of Expansion (TCE) $(°C)^{-1}$

1. It determines how the size of an object changes with the temperature changes.
2. It measures the fractional change in size per degree change in temperature at a constant pressure.

α-particle protection: the α-particles are emitted by the decay of uranium and thorium atoms contained as impurities in the packaging materials.

Pin-density: pin-count requirement has increased dramatically over the last few decades. The number of pins can be described by an empirical law, called Rent's rule.

$T = t.\ B^p$

T: number of pins

B: number of logic blocks

t: average number of terminals per logic block

P: Rent's exponent (0.5 – 0.75)

Also,

Number I/O of $= \alpha(\text{cost count})^{\beta}$

α, β constants are determined by device design.

Solved Examples

1. Estimate the number of gates that can be included on a logic gate array chip, which is to be assembled in a 100 I/O package. Assume $\alpha = 4.5$ and $\beta = 0.5$.

Solution:

No. I/O $= \alpha(\text{cost count})^{\beta}$

$100 = 4.5\ (\text{Gate count})0.5$

$= (\text{G.C})0.5$

$0.5 = 1/2$

$\therefore\ \text{G.C} = \left(\frac{100}{4.5}\right)^2 = (22.22)^2$

$\cong 493$ gates

8. A package is classified by the method used to solder the package on the printed circuit board (PCB).
   a. Pin through hole (PTH): package pins can be introduced in holes drilled in the PCB.
   b. Surface mounted technology (SMT): package pins can be directly soldered on the PCB.

9. The PTH requires an inexpensive soldering process.
   The SMT requires an expensive soldering equipment.
10. Plastic has been the dominating material for IC packages for years, although it has the disadvantage of being permeable to environmental moisture. Ceramic packages are used when power dissipation, performance, or environmental requirements justify the relative higher cost. In short, ceramic package is costlier than plastic ones.

Common IC Package Types

1. **Dual in Line Package (DIP): PTH**
   a. It is available at a lower cost.
   b. There are noise problems in high frequency applications (because of their high interconnect inductances).
2. **Pin Grid Array (PGA) Packages: PTH**
   a. The pin count is 100 ~ 400.
   b. There is higher thermal conductivity than DIP (hence, it has better power dissipation characteristics).
   c. It is costlier than DIP.
3. **Chip Carrier Packages (CCP): SMT**
   a. These are of two types: leadless and leaded chip carrier.
   b. Leadless has mechanical stress because of difference in thermal coefficient of the chip carrier and PCB.
   c. A leaded chip carrier solves this problem (as the added lead can accommodate small dimension variations caused by the difference in thermal coefficient).
4. **Quad Flat Packs (QFPs): SMT**
   a. It is similar to leaded chip carrier, except the leads extend outward rather than being bent under the package body.
   b. The pin count is up to 500 (popular in recent years).
5. **Ball Grid Array (BGA): SMT**
   a. Compared to DIPs and QFPs, it has smaller inductance and capacitance as it provides a short vertical path for the current.
   b. The pin count similar to the leaded chip carrier package, except that the leads are outward rather than bent under the package body.
6. **Multi-Chip Modules**
   a. These are used for special applications requiring high performance, where multiple chips are assembled on a common substrate contained in a single package.
   b. The advantages are significant savings in the overall system size, reduced package lead counts, and faster operation as chips can be placed in close proximity.
7. **System in Package (SIP)**
   a. It integrates multiple functions and devices, such as RF, analogue, digital memory, and MEMS in a single package.
   b. It can significantly reduce the size, weight, cost, and power consumption of a chip.
   c. It needs a faster time to market.
   d. It has a higher yield.

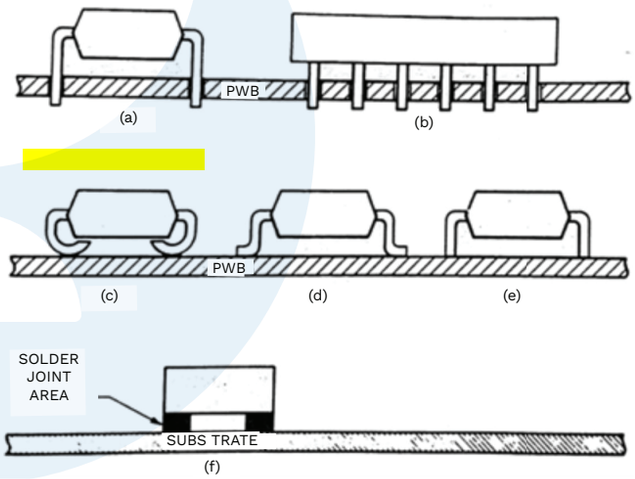
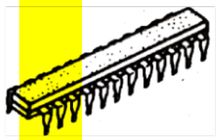


Fig. 6.1 Example of Packages and Mounting Methods to PWB in Use Today (a) and (b) Show Typical Through Hole Packages; (c) to (f) Show Typical Surface Mounted Packages (a) Dual in Line Package, (b) Pin Grid Array Package, (c) 'J' Leaded Packages, Leaded Chip Carrier, or Small Outline, (d) Gull Wing Leaded Packages, Chip Carrier, or Small Outline, (e) Butt Leaded Package and Small Outline Dual in Line Type, and (f) Leadless Type and Ceramic Chip Carrier Mounted to a Matching Ceramic Substrate

Mounting Method	Package Name	Outline	Lead Spacing
Through Hole Mounting	S-DIP (Shrink DIP)		1.78 mm
	Skinny DIP (Or Slim DIP)		2.54 mm Ceramic width 2/3 or 1/2
	SIP (Single Inline)		2.54 mm
	ZIP (Zigzag Inline)		2.54 mm
	QUIP (Quad Inline)		1.27 mm

FPO

Table 6.1 A Variety of Package Designs

Packaging Design Considerations

1. VLSI Design Rules

a. Rules must be generated for the particular package type used and must be compatible with the assembly equipment to be used.

b. These decisions should be made early, preferably before the chip layout is started and definitely before the layout is completed.

c. As the I/O grows, the device size shrinks, and the space required for interconnection could represent a major fraction of the chip area. To avoid this problem, the effective bonding pad size and spacing (also called pitch) should be reduced.

Solved Examples

2. How many bond pads placed on a 100 μm pitch can one place along a bond pad locus that is 1000 μm long?

Solution:

Let N = Number of bond pads area, the 1000 μm locus

∴ (N –1) spaces required in between bond pads each being 100 μm in length (pitch)

$(N - 1)\ 100 = 1000$

$N - 1 = \frac{1000}{100} = 10$

$N = 11$

2. Thermal Design Considerations

a. The objective is to keep the operating junction temperature of silicon die low enough to avoid temperature activated failures.

b. In a simplistic heat transfer model of a packaged die, the heat is transferred from the die to the surface of the package by conduction and from the package surface to the ambient by convection and radiation.

$\theta_{ja} = \theta_{jc} + \theta_{ca}$ (°C / watt)
$\theta_{jc} = (T_j - T_c)/P$
$\theta_{ca} = (T_c - T_a)/P$
θ_{ja} = Jn to ambient thermal resistance
θ_{jc} = Jn to case thermal resistance [case = package]
θ_{ca} = Case to ambient thermal resistance
T = Average die or Jn temperature (°C)
T_c = Average case temperature (°C)
T_a = Ambient temperature (°C)
P = Power (watts)

Generally, $\theta_{ca} \cong 0$

θ_{jc}: mainly a function of package materials and geometry

With the higher power requirements of VLSI applications, one must also consider the temperature dependence of the material selected in the design. For example, the thermal conductivity of common ceramic materials used in a packaging is temperature dependent.

θ_{ca} depends on package geometry and conditions of the ambient in the operating environment (free or forced convection heat transfer).

Solved Examples

3. What is the θ_{ja} for a device dissipating 550 mW into an ambient of 70°C and operating at a junction temperature of 125°C?

Solution:

$$\theta_{ja} = \theta_{jc} + \theta_{ca}$$

$$\theta_{jc} = \frac{T_j - T_c}{P} + \frac{T_c - T_a}{P}$$

$$= \frac{T_j - T_c + T_c - T_a}{P} = \frac{T_j - T_a}{P}$$

$$= \frac{125 - 70}{0.550} = 100°C / W$$

- Most designers use experimental methods to determine the thermal resistance of a package configuration.
- Results show that packaged devices with a preferred heat flow path (most ceramic packages) may be thermally characterised independent of external heat sinking, while those with no preferred heat flow path (most plastic packages) should be thermally characterised with a specific mounting arrangement and convection heat flow environment.

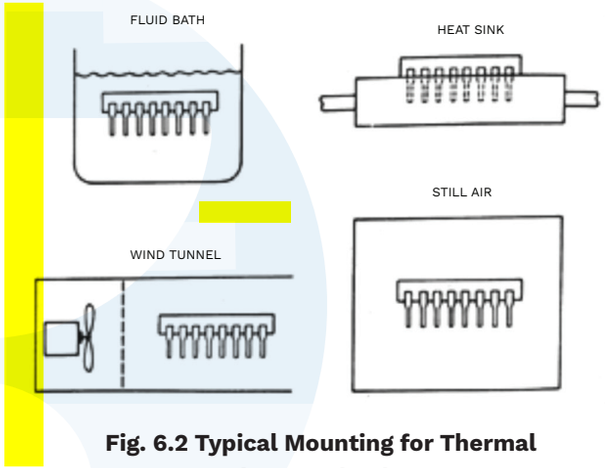


Fig. 6.2 Typical Mounting for Thermal Characterisation

3. Electrical Considerations

a. Low ground resistance (the minimum power voltages drop)

b. Short signal leads (minimum self-inductance)

c. Minimum power supply spiking due to signal lines simultaneously switching

d. Short paralleled signal runs (minimum mutual inductance and cross talk)

e. Short length signal runs near a ground plane (minimum capacitive loading)

f. Maximum use of matched impedances to avoid signal reflection

g. Lead inductances on the new SM packages are significantly lower than their TH counterparts (this is due to short leads on the leaded SM package,

no leads on leadless, and shorter lead traces within the package because of the smaller size of SM than I/O equivalent TH package)

h. The most important practical electrical design problem is noise reduction

i. When a line switches, the voltage induced in the ground line is:

$$V_i = L_g \frac{di}{dt}$$

V_i: induced voltage

L_g: inductance of the ground lead

$\frac{di}{dt} \rightarrow$ derivative of current with respect to time

If J lines are switching, then:

$$V_i = L_g \sum_j \frac{di}{dt}$$

j. To reduce V_i, multiple grounds are used to lower L_g.

k. If m ground leads are used, L_g will be L_g/m.

l. Inductance can be reduced significantly through the use of large area power and ground planes within the package.

4. Mechanical Design Considerations

a. Prefer using materials matched in physical properties in particular thermal coefficient of expansion (TCE).

b. Better matches by making engineering trade-offs that did not degrade the package device yields, performance, or reliability.

c. For example, in SSI plastic molded packages, the si die is attached directly to the lead frame (typically Alloy 42).

d. Alloy 42 minimises thermal stress.

e. In VLSI, alloy 42 could bring on higher thermal stresses.

f. Elastic module (E) and thermal conductivity k (W/cm°C) are points of concern.

g. Elastic module is the measure of the stiffness of a solid material.

$$\frac{\text{Tensile stress}}{\text{Tensile strain}} \quad \begin{array}{l} \text{Stress} = F / A \\ \text{Strain} = \text{Proportional deformation} \end{array}$$

h. Thermal stress increases with increasing E and decreasing thickness.

i. Analysis has developed sophisticated stress models and introduced computer aids, such as first element analysis (FEA) to increase their capability to characterise the thermal stresses and deformations throughout the enter plastic package with the goal of optimising the reliability of the package.

j. The FEA quickly predicts the impact of design and process changes on stress related failure mechanisms.

Practice Exercise

Level I

1. What is the preferred IC package for experimental purpose?
(A) Flat-pack
(B) DIP
(C) Metal can package
(D) Transistor pack

2. Packaging of die provides:
(A) mechanical protection
(B) environmental protection
(C) Both (A) and (B)
(D) None of the above

3. Important packaging concerns are:
(A) hermetic seals
(B) thermal conductivity
(C) pin density
(D) All of the above

4. How the packages are classified?
(A) Packages are classified by the method used to solder the package on the PCB.
(B) Packages are classified on the basis of their pin count.
(C) Packages are classified on the basis of thermal conductivity.
(D) Packages are classified on the basis of the protection level provided by the package.

5. Which of the statements is true?
(A) Pin Through Hole (PTH) is more expensive than Surface Mount Technology (SMT).
(B) PTH is less expensive than SMT.
(C) PTH is better than SMT.
(D) SMT is mostly used package type.

Level II

1. Choose the correct statement.
(A) Ceramic package is costlier than plastic ones.
(B) DIP has the maximum pin count of up to 64.
(C) Plastic has been the commonly used package material.
(D) All of the above

2. How many bond pads placed on a 100 µm pitch can one place along a bond pad locus that is 1000 µm long?
(A) 11
(B) 10
(C) 12
(D) 9

3. What is the $\ddot{Y}_{ja}$ for a device dissipating 550 mW into an ambient of 70^0 C and operating at a junction temperature of 125^0C?
(A) 110° C/W
(B) 100° C/W
(C) 120° C/W
(D) 195° C/W

4. Packaging design considerations are:
(A) VLSI design rules
(B) electrical considerations
(C) mechanical considerations
(D) All of the above

5. FEA stands for:
(A) Finite Element Analysis
(B) Fixed Element Analysis
(C) Finite Electrical Adaptation
(D) None of the above

Answer Key

Level I

1. (B) **2.** (C) **3.** (D) **4.** (A) **5.** (B)

Level II

1. (D) **2.** (A) **3.** (B) **4.** (D) **5.** (A)